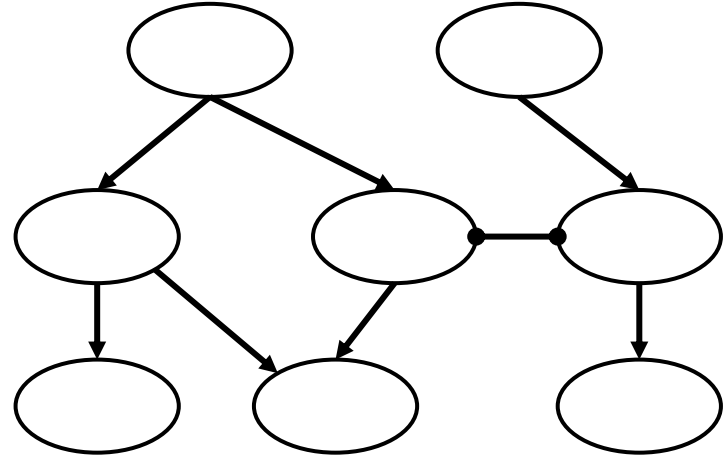




Lecture 8 (Routing 5)

Inter-Domain Routing Exercises ANS

Q1. BGP

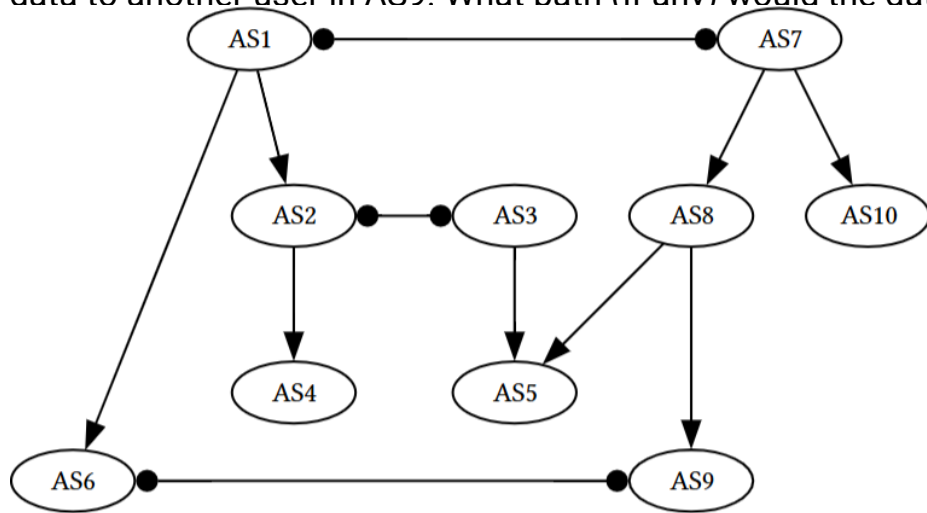
Consider the following configuration of autonomous systems and their relationships.

A square-triangle line indicates a provider-customer relationship, and a circle-circle line indicates a peer relationship. We have two tier 1 providers (AS1 and AS7). Assume we are using the following selection and export rules for all ASes:

Selection: Prefer to send data through customers > peers > providers in order to minimize costs.

Export: To customers, advertise all paths. To all other ASes, advertise only paths to customers (and paths to your own AS, of course).

- 1) A user in AS4 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?
- 2) A user in AS6 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?
- 3) A user in AS1 tries to send data to another user in AS9. What path (if any) would the data take?



Q. BGP ANS

1) A user in AS4 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?

ANS: $AS4 \rightarrow AS2 \rightarrow AS3 \rightarrow AS5$

2) A user in AS6 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?

ANS: $AS6 \rightarrow AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS5$. Since AS2 won't advertise its peer's (AS3's) path (about how to reach AS5) to its provider (AS1), so AS1 doesn't know it can reach AS5 through AS2.

3) A user in AS1 tries to send data to another user in AS9. What path (if any) would the data take?

ANS: $AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS9$. Same reasoning as Part 2).

